

CPTG and the Hubble-Tension Bridge

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Abstract

The Hubble tension compares an early-universe CMB/acoustic inference of the present expansion scale with a locally calibrated distance-ladder inference. Curvature Polarization Transport Gravity (CPTG) organizes these observations through one native cosmological π branch and two projected observational branches, acoustic and luminosity. The three descriptions share the same geometric phase $N = \ln a$ but generally accumulate different branch times through $H_b dt_b = dN$.

The native CPTG normalization and exact acoustic projection give

$$H_{0,\text{ac}}^{\text{CPTG}} = 67.4777967351 \text{ km s}^{-1} \text{ Mpc}^{-1}, \quad (1)$$

which differs from the Planck 2018 base- Λ CDM CMB inference $67.4 \pm 0.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$ by 0.156σ when the CPTG coordinate is treated as fixed. The four-anchor SH0ES distance ladder gives

$$H_0^{\text{SH0ES}} = 73.17 \pm 0.86 \text{ km s}^{-1} \text{ Mpc}^{-1}, \quad (2)$$

and conditions the present luminosity endpoint through

$$\mathcal{L}_{\text{loc}} = 1.054075942846 \pm 0.012389029805, \quad \Xi_{\text{loc}}(1) = 0.948698247775 \pm 0.011150478244. \quad (3)$$

The reciprocal quantity $\Xi_{\text{loc}}(1)$ is a present-epoch clock-rate factor, not an integrated time stretch. The acoustic and SH0ES-conditioned luminosity endpoints are separated by

$$\frac{H_{0,L}}{H_{0,\text{ac}}^{\text{CPTG}}} - 1 = 0.0843566853 = 8.43566853\%. \quad (4)$$

The bridge therefore represents the Planck and SH0ES endpoints as distinct observational projections of one native geometric history rather than forcing them onto a single branch clock. The acoustic side is derived from the CPTG branch before comparison with Planck; the luminosity endpoint is observationally conditioned by SH0ES.

1 Introduction

The conventional Hubble tension is an inconsistency between parameter inferences obtained from different observational data sets and model pipelines. A representative early-universe value is the Planck 2018 base- Λ CDM CMB inference

$$H_0^{\text{Planck}} = 67.4 \pm 0.5 \text{ km s}^{-1} \text{ Mpc}^{-1}, \quad (5)$$

while the four-anchor SH0ES distance ladder gives

$$H_0^{\text{SH0ES}} = 73.17 \pm 0.86 \text{ km s}^{-1} \text{ Mpc}^{-1} \quad (6)$$

[1, 2]. The SH0ES value combines four geometric distance anchors with homogeneous HST Cepheid photometry and the Cepheid–SN Ia distance ladder. The underlying ladder is a simultaneous three-rung construction linking geometric Cepheid calibration, Cepheids and SNe Ia in nearby hosts, and SNe Ia in the Hubble flow [3].

The direct observational separation is

$$\Delta H_0^{\text{obs}} = 73.17 - 67.4 = 5.77 \text{ km s}^{-1} \text{ Mpc}^{-1}, \quad (7)$$

with combined quoted uncertainty

$$\sigma_{\text{obs}} = \sqrt{0.5^2 + 0.86^2} \simeq 0.995 \text{ km s}^{-1} \text{ Mpc}^{-1}. \quad (8)$$

Therefore,

$$\frac{\Delta H_0^{\text{obs}}}{\sigma_{\text{obs}}} \simeq 5.80\sigma. \quad (9)$$

This arithmetic treats the two quoted uncertainties as independent and combines them in quadrature. It is distinct from a theory-to-observation residual in which a CPTG comparison coordinate is treated as fixed, and it does not remove the base- Λ CDM model dependence of the Planck inference.

CPTG [4, 5] does not identify either reported endpoint with the native geometric branch by definition. It first fixes a geometric expansion branch and then maps that branch into the coordinate system used by a specified observational comparison. The bridge therefore gives high evidentiary weight to the observed values themselves while asking whether their different observational constructions can be represented as channel-specific projections of one underlying geometry.

The central present-epoch relation is

$$H_{0,\text{ac}}^{\text{CPTG}} \longleftarrow H_0^{(\pi)} \longrightarrow H_{0,L}. \quad (10)$$

The acoustic endpoint is derived from the native branch. The luminosity endpoint is the SH0ES-conditioned observational projection. The more general branch structure is established next: one native geometric history is evaluated at a common phase while its acoustic and luminosity projections accumulate distinct branch times.

2 Native Branch, Projected Clocks, and Analytic Dependency Chain

CPTG uses a shared geometric phase

$$N \equiv \ln a. \quad (11)$$

For

$$b \in \{\pi, \text{ac}, L\}, \quad (12)$$

each branch clock is defined by

$$\frac{dN}{dt_b} = H_b(N), \quad H_b(N) dt_b = dN. \quad (13)$$

Thus equal N means equal geometric phase, not equal accumulated clock reading:

$$t_b(N) \not\equiv t_{b'}(N), \quad b \neq b'. \quad (14)$$

The luminosity projection may be written generally as

$$H_L(N) = \mathcal{L}(N)H_\pi(N), \quad \Xi_L(N) = \mathcal{L}(N)^{-1}, \quad (15)$$

so that

$$\frac{dt_L}{dt_\pi} = \Xi_L(N), \quad H_L dt_L = H_\pi dt_\pi = dN. \quad (16)$$

At the present phase $N = 0$, the SH0ES-conditioned endpoint gives $\mathcal{L}(0) = \mathcal{L}_{\text{loc}}$ and $\Xi_L(0) = \Xi_{\text{loc}}(1)$.

The native normalization used here is not selected by interpolating between the Planck and SH0ES values. It follows from the fixed CPTG geometric branch [4, 5]. The branch exponent and geometric acceleration anchor are

$$p_C = \pi, \quad (17)$$

$$a_{\pi,0} = \pi^{3/2} \times 10^{-10} \text{ m s}^{-2} = 5.568327996831708 \times 10^{-10} \text{ m s}^{-2}. \quad (18)$$

The late and transport weights are

$$\mathcal{B}_\pi = 1 - \frac{1}{\pi}, \quad \mathcal{T}_\pi = \frac{1}{\pi}. \quad (19)$$

The asymptotic expansion scale follows directly from the acceleration anchor,

$$H_\infty = \frac{a_{\pi,0}}{c} = 57.3131992073 \text{ km s}^{-1} \text{ Mpc}^{-1}, \quad (20)$$

and the present native normalization is

$$H_0^{(\pi)} = \frac{H_\infty}{\sqrt{\mathcal{B}_\pi}} = 69.4162507897 \text{ km s}^{-1} \text{ Mpc}^{-1}. \quad (21)$$

The native expansion law is therefore

$$E_\pi(a)^2 = \mathcal{B}_\pi + \mathcal{T}_\pi a^{-\pi}, \quad (22)$$

with

$$H_\pi(a) = H_0^{(\pi)} E_\pi(a). \quad (23)$$

The acoustic projection is derived from the native branch through

$$p_{\text{ac}} = 3 - \frac{\pi}{100} = 2.968584073464, \quad (24)$$

so that

$$\mathcal{G}_T = \frac{p_{\text{ac}}}{p_C} = 0.9449296585513721, \quad (25)$$

and

$$\sqrt{\mathcal{G}_T} = 0.9720749243506758. \quad (26)$$

Define the nonconstant native contribution

$$X_\pi(a) \equiv E_\pi(a)^2 - \mathcal{B}_\pi = \mathcal{T}_\pi a^{-\pi}. \quad (27)$$

The exact normalized acoustic projector is

$$X_{\text{ac}}(a) = \mathcal{T}_\pi^{1-\mathcal{G}_T} X_\pi(a)^{\mathcal{G}_T} = \mathcal{T}_\pi a^{-p_{\text{ac}}}, \quad (28)$$

which preserves the native normalization at $a = 1$. Hence

$$E_{\text{ac}}(a)^2 = \mathcal{B}_\pi + \mathcal{T}_\pi a^{-p_{\text{ac}}}, \quad (29)$$

and the acoustic comparison normalization is

$$H_{0,\text{ac}}^{\text{CPTG}} = H_0^{(\pi)} \sqrt{\mathcal{G}_T} = 67.4777967351 \text{ km s}^{-1} \text{ Mpc}^{-1}. \quad (30)$$

This acoustic normalization is the CPTG CMB-facing comparison coordinate. These relations follow from the *Large-Scale Distance and Acoustic-Ruler Comparison Layer*, including the subsection *Clock and CMB Acoustic-Density Coordinates*, of the Pi-branch construction and are fixed before the Planck or SH0ES endpoint comparison is made [5].

This dependency chain is important because neither $H_0^{(\pi)}$ nor $\sqrt{\mathcal{G}_T}$ is obtained from the Planck or SH0ES endpoint values. The observational values enter only after the native and acoustic quantities have been fixed.

3 Comparison Coordinates and Observational Status

The Hubble bridge contains native CPTG quantities, derived CPTG comparison coordinates, and external observational coordinates. The categories are distinct but are used together in the bridge.

The compact present-epoch bridge is

$$67.4778 \longleftarrow 69.4163 \longrightarrow 73.17. \quad (31)$$

The corresponding native-relative endpoint factors are

$$\frac{H_{0,\text{ac}}^{\text{CPTG}}}{H_0^{(\pi)}} = 0.972074924351, \quad \frac{H_{0,L}}{H_0^{(\pi)}} = 1.054075942846, \quad (32)$$

while the acoustic-to-luminosity endpoint separation is

$$\frac{H_{0,L}}{H_{0,\text{ac}}^{\text{CPTG}}} - 1 = 0.0843566853 = 8.43566853\%. \quad (33)$$

The left endpoint is the exact acoustic projection of the native branch. The right endpoint is the observed SH0ES coordinate expressed through the CPTG luminosity/time comparison map.

Quantity	Value	Class	Role in the bridge
$a_{\pi,0}$	$5.568327996831708 \times 10^{-10} \text{ m s}^{-2}$	Native invariant	Geometric acceleration anchor upstream of the observational comparison coordinates.
$H_0^{(\pi)}$	69.4162507897	Derived branch quantity	Derived from $a_{\pi,0}$ and $\mathcal{B}_\pi$; not set by Planck or SH0ES.
$\sqrt{\mathcal{G}_T}$	0.9720749243506758	Derived branch quantity	Derived from p_{ac}/p_C ; maps the native branch to the acoustic comparison coordinate.
$H_{0,\text{ac}}^{\text{CPTG}}$	67.4777967351	Derived comparison coordinate	Fixed acoustic/CMB projection compared with the Planck base- Λ CDM inference.
H_0^{Planck}	67.4 ± 0.5	External base- Λ CDM CMB inference	Planck 2018 model-inferred CMB/acoustic comparison value.
H_0^{SH0ES}	73.17 ± 0.86	External local distance-ladder inference	Four-anchor local distance-ladder coordinate.
$\mathcal{L}_{\text{loc}}$	$1.054075942846 \pm 0.012389029805$	Data-conditioned comparison coordinate	Luminosity endpoint factor conditioned on SH0ES and referenced to the native CPTG normalization.
$\Xi_{\text{loc}}(1)$	$0.948698247775 \pm 0.011150478244$	Data-conditioned comparison coordinate	Reciprocal present-epoch clock-rate factor conditioned on SH0ES.
$H_{0,L}$	73.17 ± 0.86	Data-conditioned comparison coordinate	SH0ES endpoint expressed through the CPTG luminosity/time projection; not an independent native prediction.

Table 1: Native, projected, data-conditioned, and observational coordinates for the Hubble-tension bridge. H_0 values are in $\text{km s}^{-1} \text{Mpc}^{-1}$. Observational uncertainties are carried explicitly on the Planck and SH0ES entries and on quantities directly inherited from the SH0ES measurement.

4 Acoustic Projection and Planck Comparison

The acoustic comparison is the geometry-fixed side of the bridge. Using only the fixed CPTG quantities,

$$H_{0,\text{ac}}^{\text{CPTG}} = 69.4162507897 \times 0.9720749243506758 = 67.4777967351 \text{ km s}^{-1} \text{Mpc}^{-1}. \quad (34)$$

Relative to the Planck 2018 base- Λ CDM CMB inference,

$$\Delta H_{0,\text{ac}} = H_{0,\text{ac}}^{\text{CPTG}} - H_0^{\text{Planck}} = 0.0777967351 \text{ km s}^{-1} \text{Mpc}^{-1}. \quad (35)$$

If the fixed CPTG quantities are treated as exact theory coordinates for this comparison and only the quoted Planck uncertainty is used, the residual is

$$\frac{\Delta H_{0,\text{ac}}}{0.5} \simeq 0.156\sigma. \quad (36)$$

No additional theory uncertainty is assigned in that quoted residual. If an uncertainty distribution is assigned to the native CPTG inputs, it must be propagated before the residual is interpreted as a combined statistical pull.

The acoustic result is therefore a direct geometry-to-observation comparison: the fixed native branch and fixed transport rule produce an H_0 -like comparison coordinate that can be compared with the Planck base- Λ CDM inference.

5 Luminosity Projection and SH0ES-Conditioned Endpoint

The luminosity projection is expressed generally through

$$H_L(N) = \mathcal{L}(N)H_\pi(N), \quad \Xi_L(N) = \mathcal{L}(N)^{-1}. \quad (37)$$

At the present phase, the SH0ES-conditioned endpoint is

$$H_{0,L} = H_0^{(\pi)} \mathcal{L}_{\text{loc}}. \quad (38)$$

Using the four-anchor SH0ES observation [2],

$$H_0^{\text{SH0ES}} = 73.17 \pm 0.86 \text{ km s}^{-1} \text{ Mpc}^{-1}, \quad (39)$$

gives

$$\mathcal{L}_{\text{loc}} = \frac{73.17}{69.4162507897} = 1.054075942846. \quad (40)$$

Treating the native CPTG normalization as a fixed comparison coordinate, the SH0ES uncertainty propagates linearly as

$$\sigma_{\mathcal{L}} = \frac{0.86}{69.4162507897} = 0.012389029805, \quad (41)$$

so that

$$\mathcal{L}_{\text{loc}} = 1.054075942846 \pm 0.012389029805. \quad (42)$$

Time is part of the same projection. Because all branches share dN , the native and luminosity clocks obey

$$\frac{dt_L}{dt_\pi} = \Xi_L(N), \quad H_L dt_L = H_\pi dt_\pi = dN. \quad (43)$$

At the present epoch this gives the exact reciprocal relation

$$\Xi_{\text{loc}}(1) = \frac{1}{\mathcal{L}_{\text{loc}}} = \frac{H_0^{(\pi)}}{H_0^{\text{SH0ES}}} = 0.948698247775. \quad (44)$$

First-order propagation of the quoted SH0ES uncertainty gives

$$\sigma_{\Xi} = 0.011150478244, \quad (45)$$

and therefore

$$\Xi_{\text{loc}}(1) = 0.948698247775 \pm 0.011150478244. \quad (46)$$

The luminosity-rate and clock-rate descriptions are reciprocal forms of the same projected branch relation, not two independent corrections.

At fixed dimensionless distance-redshift shape, the corresponding normalization-scale distance-modulus relation is

$$\Delta\mu = 5 \log_{10} \left(\frac{H_{0,\text{ref}}}{H_{0,L}} \right). \quad (47)$$

Using the native branch as the reference gives

$$\Delta\mu_{\pi \rightarrow L} = 5 \log_{10} \left(\frac{69.4162507897}{73.17} \right) = -0.1143595 \text{ mag}, \quad (48)$$

while using the fixed CPTG acoustic projection gives

$$\Delta\mu_{ac \rightarrow L} = 5 \log_{10} \left(\frac{67.4777967351}{73.17} \right) = -0.1758608 \text{ mag}. \quad (49)$$

With the theory coordinates held fixed, the SH0ES uncertainty contributes approximately

$$\sigma_{\Delta\mu} = \frac{5}{\ln 10} \frac{0.86}{73.17} = 0.0255223 \text{ mag}. \quad (50)$$

These are normalization-scale comparison coordinates. They are not asserted to be universal redshift-independent magnitude corrections.

6 Bridge Diagnostics and Comparison Trajectories

Figure 1 displays the evidentiary structure directly. The Planck and SH0ES points include their quoted observational uncertainties; the CPTG acoustic projection and native branch are fixed theory coordinates in the present comparison; the luminosity CPTG point carries the SH0ES observational uncertainty through the luminosity/time comparison map.

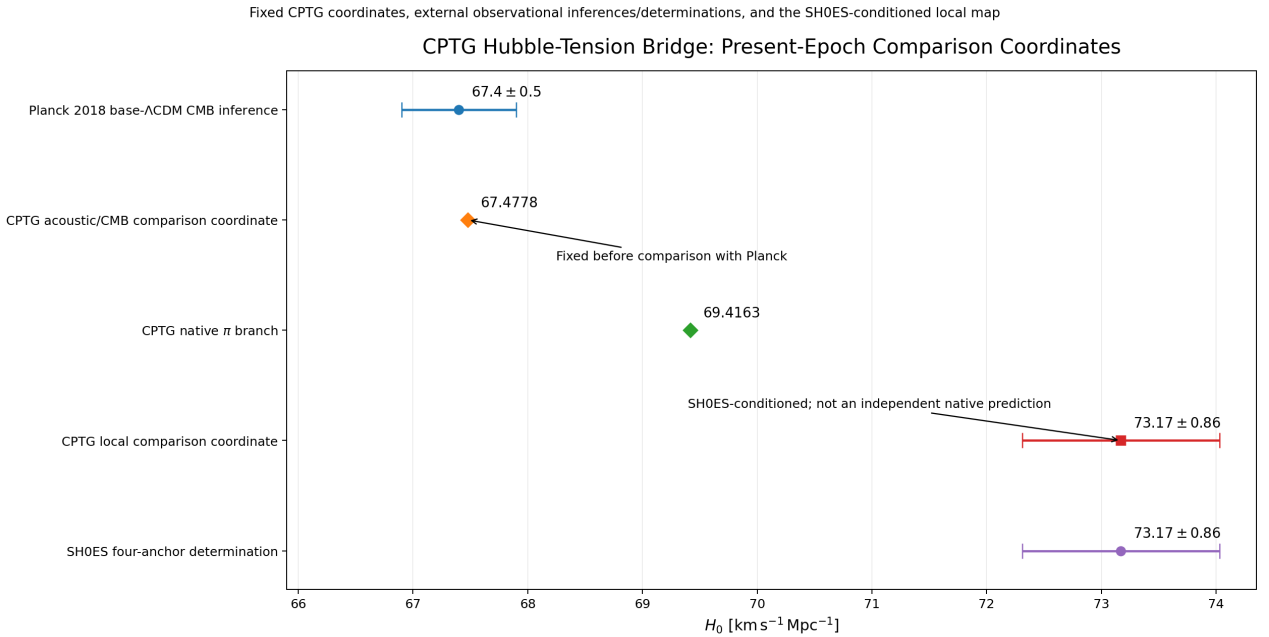


Figure 1: Present-epoch Hubble-bridge map generated directly from the fixed CPTG equations and the declared Planck and four-anchor SH0ES observational coordinates. The acoustic side is a fixed CPTG projection compared with Planck. The luminosity endpoint is the four-anchor SH0ES observation expressed on the CPTG luminosity/time projection, with the observational uncertainty retained.

A full-history comparison can be constructed without promoting the local ladder into a measured expansion history. The acoustic/CMB density-weighted comparison uses

$$\Omega_{m,ac} = \frac{p_{ac}}{3\pi} = 0.314976552850, \quad (51)$$

$$\Omega_{\Lambda,\text{ac}} = 1 - \Omega_{m,\text{ac}} = 0.685023447150. \quad (52)$$

The corresponding acoustic-density comparison background is

$$E_{\text{ac},\rho}(a)^2 = \Omega_{\Lambda,\text{ac}} + \Omega_{m,\text{ac}} a^{-p_{\text{ac}}}, \quad (53)$$

so that

$$H_{\text{ac},\rho}(a) = H_{0,\text{ac}}^{\text{CPTG}} E_{\text{ac},\rho}(a). \quad (54)$$

The term “density” identifies the acoustic/CMB density weighting of this comparison background; it does not introduce a separate native expansion branch. The symbols $\Omega_{m,\text{ac}}$ and $\Omega_{\Lambda,\text{ac}}$ are derived CPTG comparison weights fixed by p_{ac} ; they are not a new observational fit of the conventional matter and cosmological-constant density parameters.

The native geometric trajectory remains

$$H_{\pi}(a) = H_0^{(\pi)} \sqrt{\mathcal{B}_{\pi} + \mathcal{T}_{\pi} a^{-\pi}}. \quad (55)$$

For the full-history SH0ES-conditioned luminosity/time comparison realization, define

$$H_L^{\text{anchor}}(a) = H_0^{\text{SH0ES}} E_{\text{ac},\rho}(a). \quad (56)$$

The SH0ES-conditioned luminosity realization and the native branch then imply the clock map

$$\Xi_L^{\text{anchor}}(a) = \frac{H_0^{(\pi)}}{H_0^{\text{SH0ES}}} \sqrt{\frac{\mathcal{B}_{\pi} + \mathcal{T}_{\pi} a^{-\pi}}{\Omega_{\Lambda,\text{ac}} + \Omega_{m,\text{ac}} a^{-p_{\text{ac}}}}}. \quad (57)$$

The SH0ES endpoint fixes $\Xi_{\text{loc}}(1)$; the displayed $\Xi_L^{\text{anchor}}(a)$ is the SH0ES-conditioned full-history comparison associated with the declared acoustic-density trajectory. Across the SH0ES baseline Hubble-flow interval $0.0233 < z < 0.15$ [3], it varies from approximately 0.94944 to 0.95420, so the luminosity clock-rate factor remains nearly, but not exactly, constant over the comparison interval.

All dimensional ages in this section use $1 \text{ Mpc} = 3.0856775814913673 \times 10^{19} \text{ km}$ and the Julian year $1 \text{ yr} = 31,557,600 \text{ s}$.

For each displayed trajectory,

$$t(a) = \int_0^a \frac{da'}{a' H(a')}. \quad (58)$$

The acoustic-density present-age coordinate is therefore

$$t_{0,\text{ac},\rho}^{\text{CPTG}} = \frac{1}{H_{0,\text{ac}}^{\text{CPTG}}} \int_0^1 \frac{da}{a E_{\text{ac},\rho}(a)} \quad (59)$$

$$= \frac{2}{p_{\text{ac}} H_{0,\text{ac}}^{\text{CPTG}} \sqrt{\Omega_{\Lambda,\text{ac}}}} \sinh^{-1} \sqrt{\frac{\Omega_{\Lambda,\text{ac}}}{\Omega_{m,\text{ac}}}} \quad (60)$$

$$= 13.926455176722689 \text{ Gyr}. \quad (61)$$

The native geometric age remains

$$t_{0,\pi} = 12.7541210323 \text{ Gyr}, \quad (62)$$

while the SH0ES-conditioned luminosity comparison gives

$$t_{0,L}^{\text{anchor}} = \frac{H_{0,\text{ac}}^{\text{CPTG}}}{H_0^{\text{SH0ES}}} t_{0,\text{ac},\rho}^{\text{CPTG}} = 12.8430574232 \text{ Gyr}. \quad (63)$$

The last quantity is the integrated time of the SH0ES-conditioned luminosity/time comparison realization; SH0ES measures the local distance-ladder normalization rather than a cosmic age.

For clarity, the exact acoustic branch is distinct from the acoustic-density diagnostic. The exact acoustic projected background is

$$E_{\text{ac}}(a)^2 = \mathcal{B}_\pi + \mathcal{T}_\pi a^{-p_{\text{ac}}}, \quad (64)$$

with present acoustic-branch age

$$t_{0,\text{ac}}^{\text{CPTG}} = 13.88517358 \text{ Gyr}. \quad (65)$$

The exact acoustic branch is the physical projection defined earlier; $E_{\text{ac},\rho}$ is a separate density-weighted comparison diagnostic and does not define another branch clock.

The SH0ES-conditioned luminosity rate crosses the native rate at the common geometric phase

$$a_{\text{cross}} = 0.493259, \quad z_{\text{cross}} = 1.02733, \quad (66)$$

where

$$H_L^{\text{anchor}} = H_\pi \simeq 131.95 \text{ km s}^{-1} \text{ Mpc}^{-1}. \quad (67)$$

At that same phase the branch clocks read

$$t_\pi \simeq 5.0532 \text{ Gyr}, \quad t_L \simeq 5.3954 \text{ Gyr}, \quad (68)$$

so the accumulated separation reaches

$$\Delta t_{L\pi}^{\text{max}} \simeq 0.3422 \text{ Gyr}. \quad (69)$$

After the rate crossing the separation decreases, giving

$$t_{0,L}^{\text{anchor}} - t_{0,\pi} \simeq 0.08894 \text{ Gyr} \quad (70)$$

at the present phase. The branch-time mapping is therefore neither a constant rescaling nor a monotonic divergence.

The integrated values $t_{0,\pi}$, $t_{0,\text{ac}}^{\text{CPTG}}$, $t_{0,\text{ac},\rho}^{\text{CPTG}}$, and $t_{0,L}^{\text{anchor}}$ are time coordinates of the declared native, projected, or comparison backgrounds; they are not independent observational determinations of the physical age of the Universe.

The trajectory figure visualizes the rate and clock consequences of the declared comparison forms; it does not supply an additional observational validation.

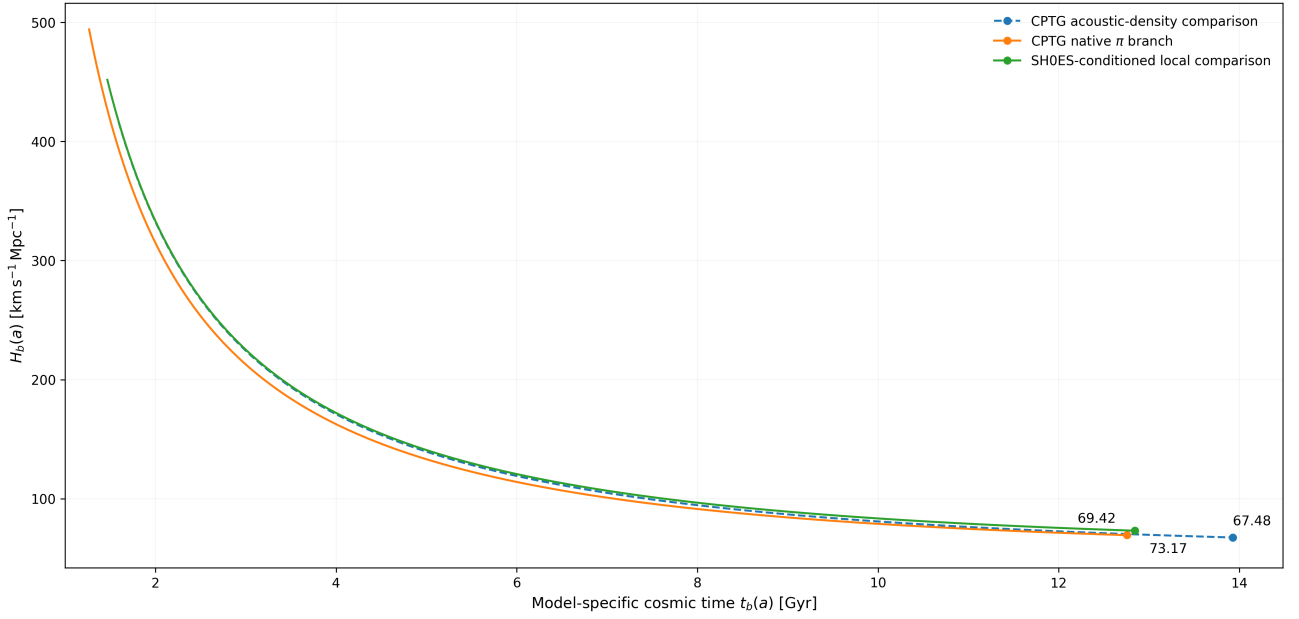


Figure 2: Native and comparison-layer trajectories generated from the equations stated in the text. The acoustic-density curve is defined by $H_{ac,\rho}(a)$. All three use the same scale-factor parameter a , but each curve is plotted against its own integrated model-specific cosmic time $t(a)$; equal horizontal positions therefore do not imply equal scale factor across curves. The displayed plotting interval is $0.20 \leq a \leq 1.00$; the analytic comparison backgrounds themselves are defined by the equations in the text. The luminosity trajectory is the SH0ES-conditioned comparison realization and is not a claim that the distance ladder directly measures a complete expansion history. The rate crossing quoted in the text occurs at a common geometric phase; because each curve is plotted against its own branch time, the crossing need not appear at the same horizontal coordinate.

7 Interpretation and External Cross-Checks

The bridge is a comparison-coordinate formulation. Planck and SH0ES are treated as observational inputs with their quoted uncertainties, while the native CPTG branch supplies the common geometric reference. The acoustic projection is fixed by the CPTG branch. The SH0ES observation conditions the present luminosity endpoint through

$$H_{0,L} = H_0^{(\pi)} \mathcal{L}_{\text{loc}}, \quad \Xi_{\text{loc}}(1) = \mathcal{L}_{\text{loc}}^{-1}. \quad (71)$$

Independent distance channels provide useful external cross-checks of the observational calibration landscape without being prerequisites for the CPTG bridge. Relevant routes include geometric maser distances, alternative standard-candle calibrations such as the tip of the red-giant branch, strong-lens time-delay distances, and controlled splits of the Cepheid–SN calibration chain by host environment, wavelength, or calibration rung. The SH0ES ladder is particularly informative because its simultaneous fit links geometric anchors, Cepheid calibration, calibrated SNe Ia, and Hubble-flow SNe Ia with their covariance structure [3].

The 2025 Chicago–Carnegie Hubble Program analysis uses independent Cepheid, TRGB, and JAGB distance routes, while the 2024 JWST analysis by Riess et al. compares Cepheid, TRGB, and JAGB subsamples after explicitly treating sample selection and common-supernova covariance [6, 7]. The CCHP paper was subsequently corrected by an erratum that revised an Appendix B slope significance but stated that the correction does not affect the rest of the Hubble analysis [8]. These channels are therefore retained as external calibration cross-checks rather than as alternate normalizations imposed on the bridge.

The comparison objective is to keep the luminosity/time coordinate explicitly tied to its observational calibration while preserving the distinction between observational conditioning and native CPTG derivation.

8 Conclusion

The CPTG Hubble bridge represents the Planck base- Λ CDM CMB inference and the four-anchor SH0ES distance-ladder determination as observational projections of one native geometric history. The shared phase is $N = \ln a$, while the native, acoustic, and luminosity branches generally accumulate different times through

$$H_b dt_b = dN. \quad (72)$$

The native normalization

$$H_0^{(\pi)} = 69.4162507897 \text{ km s}^{-1} \text{ Mpc}^{-1} \quad (73)$$

follows from the CPTG acceleration anchor and branch weight. The exact acoustic projector gives

$$H_{0,\text{ac}}^{\text{CPTG}} = 67.4777967351 \text{ km s}^{-1} \text{ Mpc}^{-1}, \quad (74)$$

which lies 0.156σ from the Planck 2018 base- Λ CDM inference 67.4 ± 0.5 when the CPTG coordinate is treated as fixed.

The four-anchor SH0ES determination

$$H_0^{\text{SH0ES}} = 73.17 \pm 0.86 \text{ km s}^{-1} \text{ Mpc}^{-1} \quad (75)$$

conditions the luminosity endpoint

$$\mathcal{L}_{\text{loc}} = 1.054075942846 \pm 0.012389029805, \quad \Xi_{\text{loc}}(1) = 0.948698247775 \pm 0.011150478244. \quad (76)$$

The acoustic and luminosity endpoints are separated by

$$\frac{H_{0,L}}{H_{0,\text{ac}}^{\text{CPTG}}} - 1 = 8.43566853\%. \quad (77)$$

The quoted Planck and SH0ES observational endpoints themselves remain separated by 5.80σ under the stated independent-error arithmetic. CPTG does not discard either result or interpolate between them to define the native branch. Instead, the acoustic side is derived from the native geometry before comparison with Planck, while the luminosity endpoint is conditioned on SH0ES.

The phase-dependent SH0ES-conditioned comparison further shows that the branch-time relation is not a constant rescaling: the native and luminosity rates cross near $z = 1.02733$, the clock separation reaches approximately 0.3422 Gyr there, and the present separation is approximately 0.08894 Gyr. The Hubble bridge is therefore a projection-and-clock statement built on one native branch, not a pair of independently normalized cosmologies.

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